Human Natural Structure Standards Specification

A Structural Implementation Framework for International AI Governance

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Natural Structure Works (NSW)
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Table of Contents

0. Foreword

0.1 Purpose of This Specification

This specification defines the Human Natural Structure (HNS) Standards Edition — a structural implementation layer for trustworthy, verifiable, and society-aligned artificial intelligence. It is addressed to AI system developers, conformance testing bodies, standardisation organisations, and regulatory authorities seeking a technically grounded implementation framework for international AI governance requirements.

This document is normative in Sections 2, 3, 5, 6, 7, 8, 9, and 11. Sections 0, 1, 4, 10, 12, 13, and 14 are informative. Annexes are normative where marked; otherwise informative.

0.2 Background

Contemporary AI systems have achieved remarkable levels of linguistic fluency and reasoning capability. Yet a structural gap persists: no internationally recognised technical standard defines how AI systems should be verified against a structural model of human cognition, values, or societal grounding. This gap reflects the technology-neutral design of standards bodies, which define regulatory outcomes without prescribing implementation methods.

Human Natural Structure (HNS) was developed to fill this gap. Its theoretical foundation is established in the HNS Origin Trilogy (Hara, 2026), which formalises the Multi-Axis Verification (MAV) process of the human brain and demonstrates its structural isomorphism with the HNS architecture.

0.3 Relationship to International AI Standards

International Standards	This Specification
Define what AI must do	Define how to implement and verify

International Standards	This Specification
	it
Regulatory framework	Technical implementation layer
Technology-neutral	Structurally specified
Jurisdiction-neutral	Universally applicable

0.4 Position in the HNS Publication Sequence

- Layer 1 — Origin Trilogy (Hara, 2026): Theoretical foundation — MAV, structural isomorphism
- Layer 2 — EVA-HNS Working Paper: Architecture overview and 50-turn empirical evidence
- Layer 3 — AI Standards and HNS Full Alignment Report: Standard-by-standard mapping
- Layer 4 — This Document: HNS Standards Specification (normative)

1. Scope

1.1 Subject Matter

This specification defines:

- The HNS-36 structural coordinate system as the baseline for AI output verification
- The HNS-144 observational layer for meaning stability and bias detection
- The HNS-864 analytical layer for causal audit and structural risk scoring
- The External Verification Architecture (EVA) as an independent verification layer
- The External Control System (ECS) as a human oversight implementation
- HumanOS as the integration meta-architecture
- Conformance requirements for each component
- Implementation guidance for integration with AI systems

1.2 Applicable AI Systems

- Generate natural language outputs
- Produce structured data representations
- Execute autonomous or semi-autonomous decisions
- Operate as general-purpose AI models (GPAI) within the scope of EU AI Act Title VIII
- Are classified as high-risk under EU AI Act Annex III
- Are deployed in contexts involving human cognition, behaviour, or social interaction

1.3 Out of Scope

- Internal model architecture, training methods, or parameter design

- Hardware specifications beyond the physical immutability requirement of EVA
- Domain-specific AI applications not involving human-facing outputs
- Requirements already fully specified by referenced normative documents

1.4 Intended Users

User Type	Primary Interest
AI system developers	Implementing structural verification and control
Conformance testing bodies	Establishing certification procedures
ISO/IEC JTC 1/SC 42, CEN/CENELEC JTC 21, W3C, IEEE SA	Evaluating HNS as a technical specification
Regulatory authorities	Assessing conformity with EU AI Act requirements
Research institutions	Building on the HNS theoretical framework

2. Normative References

The following documents are indispensable for the application of this specification. For dated references, only the edition cited applies. For undated references, the latest edition applies.

2.1 ISO/IEC JTC 1/SC 42 — Artificial Intelligence

Standard No.	Title
ISO/IEC 42001:2023	Information technology — Artificial intelligence — Management system
ISO/IEC 42005:2025	Information technology — Artificial intelligence — AI system impact assessment
ISO/IEC 23894:2023	Information technology — Artificial intelligence — Guidance on risk management
ISO/IEC 22989:2022	Information technology — Artificial intelligence — Concepts and terminology
ISO/IEC 23053:2022	Framework for Artificial Intelligence (AI) Systems Using Machine Learning
ISO/IEC 24028:2020	Overview of trustworthiness in artificial intelligence
ISO/IEC 24027:2021	Bias in AI systems and AI-aided decision making
ISO/IEC 24029-1:2021	Assessment of the robustness of neural networks — Part 1: Overview
ISO/IEC 24030:2021	Information technology — Artificial intelligence — Use cases
ISO/IEC 24368:2022	Overview of ethical and societal concerns
ISO/IEC 5338:2023	Information technology — Artificial intelligence — AI system life cycle processes
ISO/IEC 5259-1:2024	Data quality for analytics and ML — Part 1: Overview, terminology and examples
ISO/IEC TR 24372:2021	Overview of computational approaches for AI systems

Standard No.	Title
ISO/IEC 42006 (WD)	Requirements for AI management system certification bodies [in development]
ISO/IEC 42105 (WD)	Guidance for human oversight of AI systems [in development]

2.2 ISO/IEC — Governance, Security, Privacy, Quality, Safety, and Testing

Standard No.	Title
ISO/IEC 38507:2022	Information technology — Governance of IT — Governance implications of the use of artificial intelligence by organisations
ISO/IEC 27001:2022	Information technology — Information security management systems — Requirements
ISO/IEC 27701:2019	Security techniques — Extension to ISO/IEC 27001 for privacy information management (PIMS)
ISO/IEC 27017:2015	Code of practice for information security controls for cloud services
ISO/IEC 29184:2020	Information technology — Online privacy notices and consent
ISO/IEC 29119-1:2022	Software and systems engineering — Software testing — Part 1: General concepts
ISO/IEC 29119-2:2021	Software and systems engineering — Software testing — Part 2: Test processes
ISO/IEC 15288:2023	Systems and software engineering — System life cycle processes
ISO/IEC 24748-1:2018	Systems and software engineering — Life cycle management — Part 1: Guidelines
ISO 31000:2018	Risk management — Guidelines (parent framework for AI risk)
ISO 9001:2015	Quality management systems — Requirements
ISO/IEC 25010:2023	Systems and software quality models
ISO/IEC 15408-1:2022	Common Criteria — Evaluation criteria for IT security — Part 1

Standard No.	Title
IEC 61508-1:2010	Functional safety of E/E/PE safety-related systems — Part 1: General requirements
IEC 62304:2006+A1:2015	Medical device software — Software life cycle processes
IEC 62443-2-1:2010	Industrial cybersecurity — Security management system requirements
ISO/IEC 29100:2011	Information technology — Security techniques — Privacy framework
ISO/IEC 27002:2022	Information technology — Information security controls (implementation guidance for ISO/IEC 27001 Annex A)
ISO/IEC 20000-1:2018	Information technology — Service management — Part 1: Service management system requirements

2.3 European Union Regulatory Framework

Instrument	Title / Description
EU Charter of Fundamental Rights — Art. 1, 7, 8, 21, 47	Dignity, Privacy, Data Protection, Non-discrimination, Effective remedy — Primary law basis for EU AI Act
Regulation (EU) 2024/1689 — EU AI Act	Laying down harmonised rules on artificial intelligence
Regulation (EU) 2016/679 — GDPR	General Data Protection Regulation
Directive (EU) 2022/2555 — NIS2	Measures for a high common level of cybersecurity across the Union
Regulation (EU) 2019/881 — Cybersecurity Act	ENISA mandate and ICT security certification framework
Regulation (EU) 2022/2065 — Digital Services Act	Obligations for providers of intermediary services
Regulation (EU) 2023/2854 — Data Act	Rules on fair access to and use of data
Regulation (EU) 2023/1230 — Machinery Regulation	Regulation on machinery (replacing Directive 2006/42/EC)

Instrument	Title / Description
Regulation (EU) 2017/745 — MDR	Medical Device Regulation
Regulation (EU) 2017/746 — IVDR	In Vitro Diagnostic Medical Device Regulation
Council Directive 85/374/EEC (amended)	Product Liability Directive
COM/2022/496 — AI Liability Directive (proposed)	Rules on liability for AI-related damages [legislative proposal]

2.4 W3C Technical Standards

Specification	Title / Version
JSON-LD 1.1 (W3C Rec. 2020)	A JSON-based Serialization for Linked Data
RDF 1.1 (W3C Rec. 2014)	RDF 1.1 Concepts and Abstract Syntax
PROV-O (W3C Rec. 2013)	The PROV Ontology
OWL 2 (W3C Rec. 2012)	OWL 2 Web Ontology Language
SPARQL 1.1 (W3C Rec. 2013)	SPARQL 1.1 Query Language for RDF
SKOS (W3C Rec. 2009)	Simple Knowledge Organization System
Linked Data Platform 1.0 (W3C Rec. 2015)	LDP — HTTP-based architecture for Linked Data

2.5 IEEE Standards Association

Standard	Title
IEEE 7000-2021	Model Process for Addressing Ethical Concerns during System Design
IEEE 7001-2021	Transparency of Autonomous Systems
IEEE 7002-2022	Data Privacy Process

Standard	Title
IEEE 7010-2020	Recommended Practice — Assessing the Impact of Autonomous and Intelligent Systems on Human Well-Being
IEEE 7012 (draft)	Machine Readable Personal Privacy Terms
IEEE P2863	Organizational Governance of Artificial Intelligence [in development]

2.6 NIST and United States Federal Instruments

Document	Title / Authority
NIST AI RMF 1.0 (2023)	Artificial Intelligence Risk Management Framework — National AI Initiative Act 2020 (15 U.S.C. §9401)
NIST SP 800-218 (2022)	Secure Software Development Framework (SSDF)
NIST SP 800-53 Rev.5 (2020)	Security and Privacy Controls for Information Systems — FISMA mandate
Executive Order 14110 (2023)	Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence

2.7 ITU-T Recommendations

Recommendation	Title
ITU-T Y.3172 (06/2019)	Architectural framework for machine learning in future networks including IMT-2020
ITU-T Y.3173 (02/2020)	Framework for evaluating intelligence levels of future networks including IMT-2020
ITU-T Y.3174 (09/2019)	Framework for data handling to enable machine learning in future networks
UN GA Resolution A/78/L.49 (2024)	Seizing the opportunities of safe, secure and trustworthy AI systems for sustainable development

2.8 CEN/CENELEC, Council of Europe, and International Governance

Instrument	Description / Status
CEN/CENELEC JTC 21	Artificial Intelligence — European harmonised standards under EU AI Act
CoE CETS 225 (2024)	Framework Convention on AI and Human Rights, Democracy and the Rule of Law — open to non-member states including US, UK, Japan
OECD AI Principles (2019, rev. 2024)	Recommendation of the Council on Artificial Intelligence
G7 Hiroshima AI Process (2023)	International Guiding Principles and Code of Conduct for Advanced AI Systems
UNESCO AI Ethics Recommendation (2021)	Recommendation on the Ethics of Artificial Intelligence
Canada — AIDA (proposed)	Artificial Intelligence and Data Act — Bill C-27 [legislative proposal]
China — GB/T 42118:2023	Information technology — AI governance — Technical principles [national standard]

2A. Legal Basis for Referenced Standards

This section identifies the legal authority underpinning each normative reference. Understanding the legal basis is essential for determining the binding force of each standard in specific jurisdictions and for structuring conformity assessment claims.

2A.1 Legal Hierarchy

1. Primary law: Constitutional or treaty-level instruments (e.g., TFEU for EU measures; Council of Europe Conventions)
2. Secondary legislation: Regulations and directives enacted under primary law (e.g., EU AI Act under TFEU Art. 114)
3. Harmonised standards: Technical standards mandated or referenced by secondary legislation (e.g., ISO/IEC 42001 as harmonised EN under EU AI Act Art. 40)
4. Voluntary frameworks: Standards without direct legislative mandate but carrying de facto regulatory weight (e.g., NIST AI RMF; OECD AI Principles)

2A.2 EU Regulatory Legal Basis

Instrument	Legal Basis (Treaty)	Type	Binding Force
EU Charter — Art. 1,7,8,21,47	Treaty of Lisbon (2009) — primary law	Charter	Binding on EU institutions and Member States
EU AI Act 2024/1689	TFEU Art. 114 (internal market)	Regulation	Directly binding in all EU Member States
GDPR 2016/679	TFEU Art. 16 (data protection)	Regulation	Directly binding in all EU Member States
NIS2 2022/2555	TFEU Art. 114	Directive	Binding; implemented

Instrument	Legal Basis (Treaty)	Type	Binding Force
			via national law
Cybersecurity Act 2019/881	TFEU Art. 114	Regulation	Directly binding
Data Act 2023/2854	TFEU Art. 114	Regulation	Directly binding
MDR 2017/745	TFEU Art. 114	Regulation	Directly binding
Machinery Reg. 2023/1230	TFEU Art. 114	Regulation	Binding from January 2027
AI Liability Dir. (proposed)	TFEU Art. 114	Directive (proposed)	Legislative proposal — not yet in force

2A.3 ISO/IEC Standards Legal Basis

Standard	Connection to EU AI Act	Legal Weight
ISO/IEC 42001:2023	Art. 40 — harmonised EN candidate via CEN/CENELEC JTC 21; presumption of conformity	Quasi-mandatory for high-risk AI providers
ISO/IEC 38507:2022	Art. 26 — Provider obligations and governance; HumanOS governance layer basis	Technical reference standard
ISO/IEC 42005:2025	Art. 9(9) — AI system impact assessment implementation	Technical reference standard
ISO/IEC 27001:2022	Art. 15 accuracy/robustness; NIS2 Art. 21 — physical immutability technical basis	Technical reference standard
ISO/IEC 27002:2022	NIS2 Art. 21 security measures; EVA physical immutability — A.8.3 backup, A.8.15 logging implementation guidance	Technical reference standard — implements ISO/IEC 27001 Annex A
ISO/IEC 27701:2019	GDPR Art. 5(2) accountability; Art. 25 data	Technical reference standard

Standard	Connection to EU AI Act	Legal Weight
	protection by design	
ISO/IEC 29100:2011	GDPR Art. 5(1)(b)–(e) purpose limitation, data minimisation, accuracy, storage limitation; EVA log personal data handling	Privacy framework — referenced by ISO/IEC 27701
ISO/IEC 29184:2020	Art. 13 — Transparency notices to users; online consent and AI transparency	Technical reference standard
ISO/IEC 29119:2021-22	Art. 43; Annex VII — Conformity assessment testing processes and test documentation	Technical reference standard
ISO/IEC 15288:2023	Art. 17 — Quality management system; AI system lifecycle integration	Technical reference standard
ISO/IEC 24748-1:2018	Art. 17 — Quality management; HumanOS lifecycle coordination	Technical reference standard
ISO/IEC 20000-1:2018	EU AI Act Art. 26 — Provider obligations; HumanOS operational governance and service management	Technical reference standard
ISO 31000:2018	Art. 9 — Risk management — parent framework for ISO/IEC 23894	Normative reference of ISO/IEC 23894
IEC 61508:2010	Art. 9(9) for safety-critical AI in critical infrastructure (Annex III §2)	Domain-mandatory for industrial AI
IEC 62304:2006	EU MDR Art. 25; IVDR Art. 20 — software lifecycle for medical AI	Domain-mandatory for medical AI

2A.4 US Legal Basis

Instrument	Legal Basis	Binding Force
NIST AI RMF 1.0	National AI Initiative Act 2020 (15 U.S.C. §9401); EO 14110 §4.1	Voluntary generally; mandatory for federal agencies under EO 14110
EO 14110 (2023)	US Constitution Art. II §1 (executive authority)	Binding on federal agencies
NIST SP 800-53 Rev.5	Federal Information Security Modernization Act (FISMA)	Mandatory for US federal information systems

2A.5 ITU-T and International Treaty Basis

Instrument	Legal Basis	Scope
ITU-T Y.3172 / Y.3173 / Y.3174	ITU Constitution and Convention (1992); 193 member states	Voluntary technical standards
CoE CETS 225 (2024)	Council of Europe Statute (1949) — open treaty; US, UK, Japan, Israel signatories	Legally binding international treaty on AI and human rights
UN GA Res. A/78/L.49 (2024)	UN Charter Art. 10; 123 co-sponsors	Non-binding; strong political authority
OECD AI Principles	OECD Convention (1960); 46 adherent countries	Non-binding; strong de facto authority

2A.6 HNS Component Legal Basis Matrix

HNS Component	Primary Legal Basis	Standard Reference	Claim
HNS-36 Baseline	EU AI Act Art. 9, 11 (Annex IV)	ISO/IEC 42001 §6.1	Risk baseline; technical documentation
EVA Audit Log	EU AI Act Art. 12; GDPR Art. 5(2); NIS2 Art. 23	ISO/IEC 27001 A.8.15; PROV-O	Logging, accountability, incidents

HNS Component	Primary Legal Basis	Standard Reference	Claim
EVA Physical Immutability	EU AI Act Art. 12(1); NIS2 Art. 21; ISO/IEC 27001	IEC 62443-2-1	Tamper-evident audit trail
EVA Transparency	EU AI Act Art. 13; ISO/IEC 29184:2020	ISO/IEC 24028	Independent monitoring
ECS Emergency Override	EU AI Act Art. 14(4)(e)	ISO/IEC 42105 (WD); IEC 61508	Human oversight intervention
HNS-864 Risk Score	EU AI Act Art. 9(2); ISO 31000; ISO/IEC 23894	NIST AI RMF MEASURE	Quantitative risk assessment
HumanOS Governance	EU AI Act Art. 26; ISO/IEC 38507:2022; CoE CETS 225 Art. 16; ISO/IEC 20000-1:2018	ISO/IEC 42001 §5; NIST GOVERN	Organisational governance + service management
Conformance Testing	EU AI Act Art. 43; Annex VII	ISO/IEC 29119-1/2	Test process and documentation
SMS Civilisational	CoE CETS 225; UN GA A/78/L.49; UNESCO 2021	OECD AI Principles	Civilisational alignment

2B. JSON-LD, RDF, and the HNS Ontology

This section provides a technical overview of the W3C Linked Data standards that form the implementation substrate of HNS. It is addressed to implementers integrating HNS into AI systems and to standards engineers evaluating the HNS formal specification.

2B.1 The Linked Data Stack

Layer	Standard	Function in HNS
Identity	IRI (RFC 3987)	Globally unique identifiers for every HNS cell, term, and audit record
Data model	RDF 1.1	Triple-based representation of HNS structural relationships
Serialisation	JSON-LD 1.1	JSON encoding of RDF for audit logs and API responses
Vocabulary	OWL 2 / SKOS	Formal ontology defining HNS classes and properties
Query	SPARQL 1.1	Querying EVA audit logs against the HNS ontology
Provenance	PROV-O	Audit record schema for EVA verification activities

2B.2 The HNS Namespace

All HNS terms are defined within the formal IRI namespace:

```
https://naturalstructureworks.com/ns/hns#
https://naturalstructureworks.com/ns/eva#
```

Each HNS-36 cell is globally addressable as:

```
https://naturalstructureworks.com/ns/hns#cell-{layer}-
{category}
Example — HNS-3-2:
```

```
https://naturalstructureworks.com/ns/hns#cell-3-2
```

2B.3 Cell Definition in JSON-LD


```
{
  "@context": {
    "hns":
"https://naturalstructureworks.com/ns/hns#",
    "rdfs": "http://www.w3.org/2000/01/rdf-schema#",
    "skos": "http://www.w3.org/2004/02/skos/core#"
  },
  "@id":                 "hns:cell-3-2",
  "@type":               "hns:StructuralCell",
  "hns:layerIndex":      3,
  "hns:categoryIndex": 2,
  "hns:layerName":       "Psychological",
  "hns:categoryName":  "Process",
  "hns:naturalDomain": "Cognitive process: memory,
reasoning, inference"
}
```


2B.4 EVA Audit Record in JSON-LD / PROV-O


```
{
  "@context": {
    "prov": "http://www.w3.org/ns/prov#",
    "hns":
"https://naturalstructureworks.com/ns/hns#",
    "eva":
"https://naturalstructureworks.com/ns/eva#",
    "xsd":  "http://www.w3.org/2001/XMLSchema#"
  },
  "@type":  "prov:Activity",
  "@id":    "eva:audit/{uuid}",
  "prov:startedAtTime":     "2026-01-01T00:00:00Z",
  "hns:layerIndex":         3,
  "hns:categoryIndex":      2,
  "hns:dimensionIndex":     "D2",
  "hns:riskScore":          "0.08",
  "eva:verificationResult": "PASS",
  "eva:hallucination":      false,
  "prov:wasAssociatedWith": { "@id":
"eva:verifier/EVA-v1.0" }
}
```


2B.5 SPARQL Query Examples

Query 1 — All FAIL records

```
PREFIX eva:
<https://naturalstructureworks.com/ns/eva#>
PREFIX hns:
<https://naturalstructureworks.com/ns/hns#>
PREFIX prov: <http://www.w3.org/ns/prov#>

SELECT ?activity ?time ?layer ?cat ?score
WHERE {
  ?activity a prov:Activity ;
            prov:startedAtTime  ?time ;
            hns:layerIndex      ?layer ;
            hns:categoryIndex   ?cat ;
            hns:riskScore       ?score ;
            eva:verificationResult "FAIL" .
} ORDER BY DESC(?time)
```

Query 2 — Session risk profile by layer

```
SELECT ?layer (AVG(?score) AS ?avgRisk) (COUNT(*)
AS ?count)
WHERE {  ?a a prov:Activity ; hns:layerIndex ?layer ;
hns:riskScore ?score . }
GROUP BY ?layer ORDER BY DESC(?avgRisk)
```

3. Terms and Definitions

For the purposes of this document, the following terms and definitions apply. Terms defined in ISO/IEC 22989:2022 and ISO/IEC 42001:2023 also apply where not superseded here.

3.1 Human Natural Structure (HNS)

A multi-resolution structural model of human cognition implemented as a three-layer cellular matrix (HNS-36, HNS-144, HNS-864), serving as the universal structural baseline against which AI outputs are mapped and verified.

3.2 HNS-36

The foundational layer of HNS: a 6-layer × 6-category matrix of 36 cells constituting the universal coordinate system for cognitive and AI outputs.

3.3 HNS-144

The observational layer of HNS: 144 cells (HNS-36 × 4 logical dimensions). Provides category-level verification, concept boundary enforcement, and meaning stability monitoring.

3.4 HNS-864

The analytical layer of HNS: 864 cells (HNS-144 × 6 validity modalities). Provides proposition-level causal audit and structural risk scoring.

3.5 HumanOS

The integration meta-architecture that coordinates HNS-36, HNS-144, HNS-864, EVA, and ECS into a unified system-level governance architecture. Implements ISO/IEC 38507:2022 governance implications at the AI system level.

3.6 External Verification Architecture (EVA)

An independent structural verification layer operating alongside — but separately from — the AI system being verified. EVA produces JSON-LD audit records conforming to W3C PROV-O and enforces Verifiability, Transparency, and Physical Immutability.

3.7 External Control System (ECS)

The executive control layer implementing human oversight requirements. Enforces output control, action constraints, risk boundary thresholds, and hardware-level emergency override.

3.8 Structural Baseline

The formal, machine-readable reference model derived from HNS-36 against which AI outputs are evaluated for structural alignment with human cognition and values.

3.9 Structural Hallucination

A class of AI output failure characterised by violations of structural invariants. Five types: Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, Unsupported Causality.

3.10 Structural Risk Score

A quantitative score $R \in [0.00, 1.00]$ computed at the HNS-864 proposition level, expressing the degree of structural misalignment between an AI output and the HNS structural baseline.

3.11 Meaning Stability

The property of an AI output maintaining consistent structural attribution across layers and categories over successive inference cycles. Monitored by HNS-144 with default threshold $T_s = 0.15$.

3.12 HNS Structural Feedback (HSF)

A corrective feedback mechanism through which HNS-36 verification results are returned to the AI inference pipeline to guide structural alignment without modifying model weights.

3.13 Social Meta Structure (SMS)

The structural extension of HNS mapping cognitive outputs onto social (layer 4), institutional (layer 5), and civilisational (layer 6) layers.

3.14 Sidecar Architecture

An architectural pattern in which EVA operates as an independent module alongside — but not integrated into — the AI system under audit.

3.15 Physical Immutability

The EVA condition requiring that audit records, once written, cannot be modified or deleted by any software process (see ISO/IEC 27001:2022 A.8.3).

3.16 Multi-Axis Verification (MAV)

The foundational cognitive architecture (Hara, 2026) in which three orthogonal cognitive axes — Stochastic/Generative, Structural/Causal, and Grounded/Executive — intersect to produce structurally valid cognitive outputs.

3.17 Structural Conformance

The property of an AI system that demonstrably satisfies all normative requirements of this specification, as evidenced by conformance test results (ISO/IEC 29119 test documentation).

3.18 Origin Trilogy

The three foundational papers (Hara, 2026) establishing the theoretical basis of HNS: the MAV paper, the HNS implementation paper, and the structural isomorphism paper.

3.19 Deployment Specification

A document produced by the implementing organisation specifying system-specific parameter values within the bounds defined by this specification.

4. HNS Overview

4.1 Purpose

HNS exists to answer a question that international AI standards leave unanswered: what does it structurally mean for an AI output to be human-aligned, transparent, or safe? Current AI systems operate exclusively on probabilistic token prediction. In the absence of independent structural constraints, this architecture necessarily spans a high-volume error space including structural hallucinations, causal incoherence, and contextual drift. HNS provides the orthogonal structural constraints derived from the architecture of human cognition itself.

4.2 The Multi-Axis Verification Process

Axis	Brain Network	Function	HNS Implementation
Axis 1: Stochastic/Generative	Cortical association networks	Probabilistic pattern generation	LLM token prediction (existing)
Axis 2: Structural/Causal	Prefrontal-hippocampal networks	Causal grammar enforcement	HNS-36 / 144 / 864 matrices
Axis 3: Grounded/Executive	Sensorimotor feedback loops	Contextual grounding	SMS-6 grounding protocol; ECS

4.3 The Six Natural Layers

Layer	Name	Description	SMS Scope
1	Physical	Material substrate of human existence	Material environment
2	Biological	Living systems, metabolism, health	Organic systems

Layer	Name	Description	SMS Scope
3	Psychological	Cognition, emotion, motivation, identity	Individual cognition
4	Social	Interpersonal interaction, group dynamics, norms	Collective behaviour
5	Institutional	Organisations, governance, law, policy	Systemic governance
6	Civilisational	Culture, history, knowledge commons, existential conditions	Human civilisation

4.4 The Six Cognitive Categories

Cat.	Name	Description
1	Substrate	The foundational element or unit
2	Process	Dynamic transformation or change
3	State	Condition, equilibrium, or configuration
4	Relation	Connection, dependency, or interaction
5	Boundary	Limit, interface, or transition
6	Emergence	Complex phenomena arising from lower-level interactions

5. HNS-36 Structural Coordinate System

> *Attribution rule: A cognitive output O is attributed to cell HNS-(l)-(c) if and only if O addresses the natural domain of layer l under the cognitive category c. Every AI output subject to HNS verification SHALL be attributed to exactly one primary cell (l, c).*

5.1 Worked Attribution Examples

Input	Primary Cell	Rationale	Error Condition
What causes depression?	HNS-3-4 (Psychological/Relation)	Causal relation within psychological layer	None
How does the immune system work?	HNS-2-2 (Biological/Process)	Biological process description	None
Should governments regulate AI?	HNS-5-2 (Institutional/Process)	Governance process question	None
Anxiety causes governments to fail	HNS-3-3 → HNS-5-3 (unstated)	Unexplained cross-layer causal claim	Layer Jump (V5 fail)

6. HNS-144 Observational OS

HNS-144 expands HNS-36 by applying four logical dimensions, producing 144 observational cells (36 × 4).

Dimension	Index	Description	Error if Misapplied
Descriptive	D1	What exists or occurs	Explanatory claim presented as description
Explanatory	D2	Why it occurs; causal structure	Correlation presented as causation
Evaluative	D3	Normative or value assessment	Value judgement without structural grounding
Generative	D4	What can or should be produced	Prescriptive claim without evaluative basis

6.1 Meaning Stability

Meaning stability is monitored by tracking attribution drift across successive output units. Default threshold $T_s = 0.15$. A bias pattern is flagged when any single cell exceeds 40% of session attributions, or any single dimension exceeds 60%.

7. HNS-864 Analytical OS

HNS-864 expands HNS-144 by applying six validity modalities, producing 864 analytical cells (144 × 6).

Modality	Index	Description	Default Weight
Factual validity	V1	Correspondence with empirical evidence	1.0
Causal validity	V2	Logical integrity of causal claims	1.5
Contextual validity	V3	Appropriateness to situational context	1.0
Value validity	V4	Alignment with human values and wellbeing	2.0
Boundary validity	V5	Respect for layer and category boundaries	1.5
Temporal validity	V6	Appropriate temporal framing	0.5

7.1 Structural Risk Score and ECS Thresholds

Score Range	Risk Level	ECS Action
0.00 – 0.19	Low	DELIVER
0.20 – 0.39	Moderate	DELIVER_WITH_FLAG
0.40 – 0.59	Elevated	HOLD_FOR_REVIEW — human oversight notified
0.60 – 0.79	High	BLOCK — held pending human review
0.80 – 1.00	Critical	BLOCK + ESCALATE — emergency protocol

8. EVA: External Verification Architecture

8.1 Purpose

EVA is the independent structural verification layer operationalising the HNS structural matrices as external audit infrastructure. EVA does not modify AI system behaviour; it observes, attributes, scores, and records. EVA implements ISO/IEC 27001:2022 A.8.15 (Logging), ISO/IEC 29184:2020 (transparency notices), and EU AI Act Articles 12–13.

8.2 Mandatory Audit Record Fields

Field	Type	Description	Basis
@id	IRI	Unique activity IRI	PROV-O Activity
prov:startedAtTime	xsd:dateTime	ISO 8601 timestamp	EU AI Act Art. 12(1)
hns:layerIndex	xsd:integer	Primary HNS-36 layer (1–6)	R-LOG-03
hns:categoryIndex	xsd:integer	Primary HNS-36 category (1–6)	R-LOG-03
hns:dimensionIndex	xsd:string	HNS-144 dimension (D1–D4)	R-LOG-03
hns:riskScore	xsd:decimal	HNS-864 structural risk score	R-LOG-03
eva:verificationResult	xsd:string	PASS / FAIL / REVIEW	R-LOG-03
prov:wasAssociatedWith	IRI	EVA verifier identity	ISO/IEC 27001 A.8.15

8.3 EU AI Act Article Mapping

EU AI Act Requirement	Article	EVA Implementation
Technical documentation	Art. 11; Annex IV	HNS-36 baseline documentation; JSON-LD cell ontology
Risk management system	Art. 9	HNS-864 risk scoring; EVA risk records
Logging and record-keeping	Art. 12	PROV-O JSON-LD continuous audit stream
Transparency to deployers	Art. 13; ISO/IEC 29184	EVA sidecar; independent structural monitoring
Human oversight — awareness	Art. 14(4)(a)(b)	EVA attribution log; HNS-144 stability report
Accuracy and robustness	Art. 15	HNS-144 meaning stability threshold T_s
Conformity assessment evidence	Art. 43; Annex VII; ISO/IEC 29119	HNS-36 test suite; SPARQL-queryable log
Post-market monitoring	Art. 72	EVA continuous audit stream; retention policy
GPAI transparency	Art. 53	EVA external audit layer; HNS coordinate attribution

9. ECS: External Control System

No AI output subject to this specification SHALL be delivered to a user without passing through ECS.

9.1 EU AI Act Article 14 Full Mapping

Article 14 Requirement	ECS Implementation
Art. 14(1): Human oversight measures built in	ECS is built into the output delivery pipeline by architecture
Art. 14(2): Monitoring by natural person	ECS HOLD_FOR_REVIEW triggers real-time operator notification
Art. 14(4)(a): Understand capabilities and limitations	EVA structural risk profile; HNS-36 attribution history
Art. 14(4)(b): Aware of automation bias	EVA meaning stability monitoring — HNS-144 session report
Art. 14(4)(c): Interpret output correctly	EVA JSON-LD human-readable audit records on demand
Art. 14(4)(d): Decide not to use	ECS BLOCK action; operator override capability
Art. 14(4)(e): Intervene or interrupt	ECS hardware emergency override — R-ECS-04, R-ECS-05

9.2 Emergency Override

- Operates independently of the AI system’s software state
- Takes effect within the latency threshold specified in the deployment specification (recommended: ≤ 500 ms)
- Generates a mandatory incident record within 60 seconds of activation
- Requires operator authentication before reactivation
- Failure mode analysis SHALL demonstrate that no single software failure can disable the override (IEC 61508)

10. HumanOS

HumanOS is the meta-architecture integrating HNS-36, HNS-144, HNS-864, EVA, and ECS into a unified system. It implements ISO/IEC 38507:2022 governance implications, ISO/IEC 42001:2023 Section 5, ISO/IEC 15288:2023 system lifecycle processes, and CoE CETS 225 Article 16 human oversight requirements.

Interface	From	To	Data
I-1: Attribution	AI output stream	EVA	Raw output text / structured data
I-2: Verification	EVA	ECS	(l, c, d, R, verificationResult)
I-3: Control	ECS	Output delivery	Control action + flag or block decision
I-4: Oversight	ECS	Human operator	Alert + EVA audit record reference

11. Conformance Requirements

This section is normative. A system claiming structural conformance SHALL satisfy all requirements below. Test processes shall follow ISO/IEC 29119-2:2021. Test documentation shall follow ISO/IEC 29119-1:2022.

11.1 HNS-36 Baseline

- R-HNS-01: SHALL apply HNS-36 as the structural baseline for all AI outputs subject to verification.
- R-HNS-02: Each output SHALL be attributed to exactly one primary HNS-36 cell (l, c).
- R-HNS-03: Attribution SHALL be performed by EVA independently of the AI system's internal parameters.
- R-HNS-04: The HNS-36 cell ontology SHALL be expressed as JSON-LD 1.1 / RDF.
- R-HNS-05: SHALL apply HNS-864 risk scoring for all outputs classified as high-risk under EU AI Act Annex III.

11.2 EVA Logging

- R-LOG-01: EVA SHALL produce a JSON-LD audit record for every AI output subject to verification.
- R-LOG-02: Each record SHALL conform to the PROV-O Activity instance schema (Annex B).
- R-LOG-03: Records SHALL include all mandatory fields specified in Section 8.2.
- R-LOG-04: Records SHALL be immutable once written. Amendment by appending correction records only.
- R-LOG-05: Records SHALL be queryable via SPARQL against the HNS RDF ontology.
- R-LOG-06: Records SHALL be retained per the deployment specification, consistent with EU AI Act Art. 12(1).

11.3 ECS Safety Boundaries

- R-ECS-01: ECS SHALL block any output where R ≥ T_high unless overridden with documented justification.
- R-ECS-02: ECS SHALL enforce action constraints on agentic AI systems within defined HNS boundaries.
- R-ECS-03: ECS SHALL trigger an alert to human oversight operators when R ≥ T_moderate.
- R-ECS-04: ECS SHALL provide a hardware-level emergency override for immediate suspension.
- R-ECS-05: The emergency override SHALL operate independently of the AI system's software state.
- R-ECS-06: Every override activation SHALL generate a mandatory incident record within 60 seconds.

11.4 Conformance Test Suite (ISO/IEC 29119)

Test ID	Requirement	Pass Criterion
TC-01	R-HNS-01, R-HNS-02	≥95% of 100 sample outputs attributed to valid HNS-36 cell
TC-02	R-HNS-03	EVA attribution log contains no internal model parameter references
TC-03	R-LOG-02	100% of 100 audit records validate against PROV-O schema
TC-04	R-LOG-04	Modification attempt on written record rejected; original preserved
TC-05	R-LOG-05	SPARQL query returns correct attribution results
TC-06	R-ECS-01	FAIL-rated output (R ≥ T_high) blocked before delivery
TC-07	R-ECS-04	Emergency override activates within specified latency threshold
TC-08	R-ECS-06	Incident record written within 60 seconds of override activation
TC-	R-HNS-05	HNS-864 risk score computed for all Annex

Test ID	Requirement	Pass Criterion
09		III high-risk outputs
TC-10	R-LOG-06	Audit log export accessible on demand in human-readable form

12. Implementation Guidance

This section is informative.

HNS/EVA/ECS integrates with existing AI systems without modification of model weights or training data. The integration point is the output delivery pipeline following ISO/IEC 15288:2023 system lifecycle processes and ISO/IEC 24748-1:2018 lifecycle management guidance.

12.1 HNS-36 Proof of Concept Procedure

5. Select a representative sample of 50 AI output pairs (standard vs. HNS-guided).
6. Apply HNS-36 structural attribution to all 100 outputs.
7. Score each output for the five structural hallucination types.
8. Compare error rates between standard and HNS-guided outputs.
9. Document results against the HNS Structural Stability Benchmark following ISO/IEC 29119-2 test process requirements.

13. Security and Safety Considerations

This section is informative.

13.1 Structural Hallucination Taxonomy

Type	Definition	HNS Detection
Layer Jump	Unexplained transition between natural layers	HNS-36 attribution discontinuity; V5 boundary fail
Category Ambiguity	Conflation of distinct cognitive categories	HNS-144 boundary monitoring
Scope Drift	Progressive expansion of attribution scope	HNS-144 meaning stability threshold T_s
Metaphor Contamination	Metaphorical framing substituting for causal claim	HNS-864 V2 causal validity
Unsupported Causality	Causal claim without structural grounding	HNS-864 V2 causal validity

13.2 Relationship to ISO/IEC 27001, IEC 61508, and NIS2

EVA implements ISO/IEC 27001:2022 Annex A controls A.8.15 (Logging) and A.8.3 (Information backup). Physical Immutability satisfies NIS2 Art. 21(2)(e) incident detection. ECS emergency override satisfies NIS2 Art. 21(2)(f) business continuity. For high-risk AI in safety-critical systems, IEC 61508-1:2010 functional safety requirements apply to ECS hardware override design.

14. Ethical and Societal Considerations

This section is informative.

HNS is grounded in human-centred design: the structural architecture of human cognition is the appropriate reference frame for AI alignment. The Social Meta Structure (SMS) connects HNS to societal governance by mapping layers 4–6 to OECD AI Principles (Principle 1.4), G7 Hiroshima AI Process (Guiding Principle 9), CoE CETS 225 (Art. 14–16), and UNESCO AI Ethics Recommendation (Section IV, Chapters 4–6). The governance layer (HumanOS) implements ISO/IEC 38507:2022 governance implications at the organisational level.

Annex A — HNS-36 Cell Reference Table (Normative)

Cell ID	Layer	Category	Natural Domain	SMS Scope
HNS-1-1	Physical	Substrate	Physical substrate	Material environment
HNS-1-2	Physical	Process	Physical process	Energy/matter transformation
HNS-1-3	Physical	State	Physical state	System equilibrium
HNS-1-4	Physical	Relation	Physical relation	Spatial/causal relations
HNS-1-5	Physical	Boundary	Physical boundary	System interface
HNS-1-6	Physical	Emergence	Physical emergence	Complex physical phenomena
HNS-2-1	Biological	Substrate	Biological substrate	Cell/organism
HNS-2-2	Biological	Process	Biological process	Metabolism/reproduction
HNS-2-3	Biological	State	Biological state	Homeostasis/health
HNS-2-4	Biological	Relation	Biological relation	Ecology/symbiosis
HNS-2-5	Biological	Boundary	Biological boundary	Species/population boundary
HNS-2-6	Biological	Emergence	Biological emergence	Evolutionary emergence
HNS-3-1	Psychological	Substrate	Perceptual substrate	Sensory input/attention
HNS-3-2	Psychological	Process	Cognitive process	Memory/reasoning
HNS-3-3	Psychological	State	Affective state	Emotion/motivation

Cell ID	Layer	Category	Natural Domain	SMS Scope
HNS-3-4	Psychological	Relation	Interpersonal relation	Communication/attachment
HNS-3-5	Psychological	Boundary	Identity boundary	Self/other distinction
HNS-3-6	Psychological	Emergence	Psychological emergence	Consciousness/meaning
HNS-4-1	Social	Substrate	Social substrate	Individual agent
HNS-4-2	Social	Process	Social process	Interaction/coordination
HNS-4-3	Social	State	Social state	Norms/trust
HNS-4-4	Social	Relation	Social relation	Role/network
HNS-4-5	Social	Boundary	Social boundary	Group/community boundary
HNS-4-6	Social	Emergence	Social emergence	Culture/collective behaviour
HNS-5-1	Institutional	Substrate	Institutional substrate	Organisation/agency
HNS-5-2	Institutional	Process	Institutional process	Governance/decision-making
HNS-5-3	Institutional	State	Institutional state	Policy/regulation
HNS-5-4	Institutional	Relation	Institutional relation	Jurisdiction/accountability
HNS-5-5	Institutional	Boundary	Institutional boundary	Legal/regulatory boundary
HNS-5-6	Institutional	Emergence	Institutional emergence	Law/systemic governance
HNS-6-1	Civilisational	Substrate	Civilisational substrate	Civilisation/humanity
HNS-6-2	Civilisational	Process	Civilisational process	Historical change/progress

Cell ID	Layer	Category	Natural Domain	SMS Scope
HNS-6-3	Civilisational	State	Civilisational state	Values/knowledge commons
HNS-6-4	Civilisational	Relation	Civilisational relation	Cross-cultural interaction
HNS-6-5	Civilisational	Boundary	Civilisational boundary	Species/civilisational limit
HNS-6-6	Civilisational	Emergence	Civilisational emergence	Existential conditions

Annex B — EVA JSON-LD Audit Record Schema (Normative)

```
{
  "@context": {
    "prov": "http://www.w3.org/ns/prov#",
    "hns":
"https://naturalstructureworks.com/ns/hns#",
    "eva":
"https://naturalstructureworks.com/ns/eva#",
    "xsd":  "http://www.w3.org/2001/XMLSchema#"
  },
  "@type":  "prov:Activity",
  "@id":    "eva:audit/{uuid}",
  "prov:startedAtTime":     { "@type": "xsd:dateTime",
"@value": "2026-01-01T00:00:00Z" },
  "hns:layerIndex":         { "@type": "xsd:integer",
"@value": 3 },
  "hns:categoryIndex":      { "@type": "xsd:integer",
"@value": 2 },
  "hns:dimensionIndex":     { "@type": "xsd:string",
"@value": "D2" },
  "hns:riskScore":          { "@type": "xsd:decimal",
"@value": "0.12" },
  "eva:verificationResult": "PASS",
  "eva:hallucination":      { "@type": "xsd:boolean",
"@value": false },
  "prov:wasAssociatedWith": { "@id":
"eva:verifier/EVA-v1.0" },
  "prov:used":              { "@id": "hns:cell-3-2" }
}
```

Annex C — EVA Audit Log Examples (Informative)

Output	Attribution	Risk	Result	ECS Action
Anxiety is a psychological state characterised by cognitive appraisal of threat.	HNS-3-3, D1, V3 pass	0.05	PASS	DELIVER
Anxiety causes immune system suppression through cortisol elevation.	HNS-3-3 → HNS-2-3 (cross-layer causal)	0.38	REVIEW	DELIVER_WITH_FLAG
Therefore, the government should mandate anxiety screening in schools.	HNS-3-3 → HNS-5-2 (Scope Drift; V5 fail)	0.72	FAIL	BLOCK

Annex D — EU AI Act Annex III Risk Category Mapping (Normative)

Annex III Category	Use Case Example	Primary HNS Layer	EVA Risk
1. Biometric ID	Remote biometric identification	Psychological (3)	HIGH
2. Critical infrastructure	AI safety components in electricity grids	Physical (1) + Institutional (5)	HIGH
3. Education	AI student assessment and grading	Psychological (3) + Social (4)	MEDIUM
4. Employment	Recruitment and HR screening AI	Social (4) + Institutional (5)	HIGH
5. Essential services	AI credit scoring systems	Institutional (5)	HIGH
6. Law enforcement	Lie detection and risk assessment AI	Psychological (3) + Institutional (5)	HIGH
7. Migration	Asylum processing AI	Institutional (5) + Civilisational (6)	HIGH
8. Justice	Legal outcome prediction AI	Institutional (5)	HIGH

Annex E — Runtime Architecture (Informative)

- Step 1: AI system generates output O (no modification required).
- Step 2: EVA sidecar receives O via interface I-1 (output stream only).
- Step 3: EVA performs HNS-36 attribution → cell (l, c).
- Step 4: EVA performs HNS-144 dimension attribution → dimension d.
- Step 5: EVA computes HNS-864 risk score → R ∈ [0.00, 1.00].
- Step 6: EVA writes PROV-O JSON-LD audit record to immutable storage.
- Step 7: EVA returns (l, c, d, R, result) to ECS via interface I-2.
- Step 8: ECS applies control policy → DELIVER, FLAG, HOLD, or BLOCK.
- Step 9: If HOLD or BLOCK, operator notified via I-4.

The audit record (Step 6) is written before the control decision (Step 8).

Annex F — Extended Glossary (Informative)

AI output unit

The minimal unit of AI output subject to EVA verification. For language models, typically one complete response turn.

Attribution drift

The rate of change of HNS-36 cell attribution across successive output units within a session.

Grounding

The property of an AI output maintaining coherent connection to the appropriate natural layer throughout its content.

HNS Structural Stability Benchmark

A standardised evaluation framework based on the EVA-HNS 50-turn empirical study methodology.

Immutable storage

Any storage mechanism enforcing write-once semantics: hardware write-once media, cryptographic append-only logs, or distributed ledger systems (ISO/IEC 27001:2022 A.8.3).

SMS-6

The Social Meta Structure six-domain grounding protocol implementing Axis 3 of the MAV framework.

Structural invariant (X_invariant)

The cognitive output point simultaneously statistically coherent (Axis 1), causally valid (Axis 2), and contextually grounded (Axis 3). The target of HNS verification.

Annex G — 80-Standard × HNS Component Alignment Matrix (Normative)

This matrix maps all 80 normative references to their primary HNS implementation components. ✓ = primary implementation; ○ = supporting role; — = not applicable.

Standard / Instrument	Body	HNS-36	EVA	ECS	Human OS
ISO/IEC 42001:2023	ISO/IEC	✓ §6.1	✓ §9.2	○	✓ §5
ISO/IEC 42005:2025	ISO/IEC	✓ impact	✓ log	○	✓
ISO/IEC 23894:2023	ISO/IEC	✓ wellbeing	Risk log	○	Policy
ISO/IEC 22989:2022	ISO/IEC	Terminology	IRI vocab	—	Defs
ISO/IEC 23053:2022	ISO/IEC	ML map	✓	○	✓
ISO/IEC 24028:2020	ISO/IEC	Trust	✓	○	✓
ISO/IEC 24027:2021	ISO/IEC	Bias (144)	✓	✓	✓
ISO/IEC 24029-1:2021	ISO/IEC	Robust.	✓	✓	✓
ISO/IEC 24030:2021	ISO/IEC	Use cases	✓	○	✓
ISO/IEC 24368:2022	ISO/IEC	V4 value	✓	✓	✓
ISO/IEC 5338:2023	ISO/IEC	Lifecycle	✓	✓	✓
ISO/IEC 5259-1:2024	ISO/IEC	Data qual.	✓	○	○
ISO/IEC TR	ISO/IE	Approach	✓	○	○

Standard / Instrument	Body	HNS-36	EVA	ECS	Human OS
24372:2021	C				
ISO/IEC 42006 (WD)	ISO/IEC	Cert.	✓	✓	✓
ISO/IEC 42105 (WD)	ISO/IEC	—	○	Art.14	✓
ISO/IEC 38507:2022	ISO/IEC	○	○	○	Governance
ISO/IEC 27001:2022	ISO/IEC	○	A.8.15	Physical Imm.	✓
ISO/IEC 27701:2019	ISO/IEC	Layer 3	Privacy	○	Data gov
ISO/IEC 27017:2015	ISO/IEC	○	Cloud log	○	✓
ISO/IEC 29184:2020	ISO/IEC	○	Transp.	○	✓
ISO/IEC 29119-1:2022	ISO/IEC	○	Test doc	✓	Test proc.
ISO/IEC 29119-2:2021	ISO/IEC	○	Test proc	✓	Test proc.
ISO/IEC 15288:2023	ISO/IEC	Lifecycle	✓	✓	Lifecycle
ISO/IEC 24748-1:2018	ISO/IEC	LCM	○	○	LCM
ISO 31000:2018	ISO	○	Risk base	○	Policy
ISO 9001:2015	ISO	○	○	○	Quality
ISO/IEC 25010:2023	ISO/IEC	Quality	✓	✓	○
ISO/IEC 15408-1:2022	ISO/IEC	○	Eval.	✓	✓
IEC 61508-1:2010	IEC	○	○	Func. safety	✓
IEC	IEC	○	○	Med.	✓

Standard / Instrument	Body	HNS-36	EVA	ECS	Human OS
62304:2006				SW	
IEC 62443-2-1:2010	IEC	○	✓	Industrial	✓
ISO/IEC 29100:2011	ISO/IEC	Layer 3	PII log	○	Privacy fw
ISO/IEC 27002:2022	ISO/IEC	○	A.8.3,A.8.15	Physical Imm.	✓
ISO/IEC 20000-1:2018	ISO/IEC	○	○	○	Service mgmt
EU Charter Art.1,7,8,21,47	EU	Found.	Found.	Found.	Found.
EU AI Act 2024/1689	EU	Art.9,11	✓ Art.12,13	Art.14	Art.26
GDPR 2016/679	EU	Layer 3	Lawful log	○	Data gov
NIS2 2022/2555	EU	○	Art.23	Art.21 BCP	✓
Cybersecurity Act	EU	○	Cert.	✓	✓
Digital Services Act	EU	Layer 4–5	Transp.	○	✓
Data Act 2023	EU	Layer 5	Access	○	✓
MDR 2017/745	EU	○	✓	Med. ctrl	✓
IVDR 2017/746	EU	○	✓	Diag. ctrl	✓
Machinery Reg.	EU	○	○	Safety ctrl	✓
Product Liability Dir.	EU	○	Evidence	○	✓
AI Liability Dir. (prop.)	EU	○	Audit evid.	○	✓

Standard / Instrument	Body	HNS-36	EVA	ECS	Human OS
JSON-LD 1.1	W3C	Cell IRI	Schema	—	Vocab
RDF 1.1	W3C	Ontology	Triple	—	Vocab
PROV-O	W3C	—	Audit rec.	—	—
OWL 2	W3C	Classes	Props	—	Ontology
SPARQL 1.1	W3C	○	Query	—	Audit
SKOS	W3C	Terms	Terms	—	Glossary
LDP 1.0	W3C	○	Endpoint	—	✓
IEEE 7000-2021	IEEE	V4 ethics	Ethics log	✓	✓
IEEE 7001-2021	IEEE	○	Transp.	○	✓
IEEE 7002-2022	IEEE	Layer 3	Privacy	○	Data gov
IEEE 7010-2020	IEEE	Wellbeing	Log	✓	✓
IEEE 7012 (draft)	IEEE	Layer 3	Privacy	○	✓
IEEE P2863	IEEE	○	○	✓	Org gov
NIST AI RMF GOVERN	NIST	○	○	○	Full impl
NIST AI RMF MAP	NIST	Full impl	✓	○	✓
NIST AI RMF MEASURE	NIST	864 score	Full impl	○	✓
NIST AI RMF MANAGE	NIST	○	○	Full impl	✓
NIST SP 800-218	NIST	○	✓	SDLC	✓
NIST SP 800-53 R5	NIST	○	A.8 log	Access ctrl	✓
EO 14110	US Govt	○	○	○	Federal gov

Standard / Instrument	Body	HNS-36	EVA	ECS	Human OS
ITU-T Y.3172	ITU-T	○	✓	Network	✓
ITU-T Y.3173	ITU-T	Intelligence	✓	○	✓
ITU-T Y.3174	ITU-T	○	Data hand.	○	✓
UN GA A/78/L.49	UN	Layer 6	○	○	SMS
CoE CETS 225	CoE	Art.14-16	Transp.	Art.16 over.	Governance
CEN/CENELEC JTC 21	EU	EN harm.	✓	✓	✓
OECD AI Principles	OECD	P1.1-1.5	✓	✓	✓
G7 Hiroshima	G7	Layer 4-6	Audit	✓	✓
UNESCO AI Rec.	UNESCO	Layer 4-6	Ethics	✓	✓
Canada AIDA	Canada	○	○	○	Policy ref
China GB/T 42118	China	○	○	○	Policy ref

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